Linear Algebra I: Week 7

Transposes, permutations, vector spaces and subspaces

Fall 2026

Reading and objectives

Syllabus: transposes and permutations; vector spaces and subspaces.

Strang, *Introduction to Linear Algebra*, 4th ed., §2.7 (pp. 108–120) and §3.1 (pp. 121–132).

Objectives: use transpose identities; compute row/column permutations; understand $PA = LU$; recognize and construct subspaces.

Week 6 review: elimination

Gaussian elimination gives $E_k \cdots E_1 A = U$. If no exchanges occur, $A = LU$ and L stores elimination multipliers.

Example: $A = \begin{pmatrix} 2 & 1 \\ 4 & 5 \end{pmatrix}$; $R_2 \leftarrow R_2 - 2R_1$ gives $U = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and $L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$.

Week 6 review: LDU

Extract pivots from U: $U = D\widehat{U}$, where $D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and $\widehat{U} = \begin{pmatrix} 1 & 1/2 \\ 0 & 1 \end{pmatrix}$.

Thus $A = LD\widehat{U}$. The diagonal of D contains the pivots after elimination.

Week 6 review: row exchanges

$A = \begin{pmatrix} 0 & 2 \\ 3 & 4 \end{pmatrix}$ has a zero first pivot. Swap rows with $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Then $PA = \begin{pmatrix} 3 & 4 \\ 0 & 2 \end{pmatrix}$; in general the factorization is $PA = LU$.

Definition of transpose

If A is $m \times n$, then A^T is $n \times m$ with $(A^T)_{ij} = a_{ji}$.

$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ gives $A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$. Columns become rows.

Basic transpose identities

$(A^T)^T = A$, $(A+B)^T = A^T + B^T$, $(cA)^T = cA^T$.

Entrywise proof: $((A+B)^T)_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A^T + B^T)_{ij}$.

Transpose of a product

$$\boxed{(AB)^T = B^T A^T}$$

Proof: $((AB)^T)_{ij} = \sum_k a_{jk} b_{ki} = \sum_k (B^T)_{ik} (A^T)_{kj}$.

The order reverses; $A^T B^T$ may not even be defined.

Check the product rule numerically

$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}$. Then $AB = \begin{pmatrix} 8 & 2 \\ 3 & 1 \end{pmatrix}$ and $(AB)^T = \begin{pmatrix} 8 & 3 \\ 2 & 1 \end{pmatrix}$.

$B^T A^T = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 3 \\ 2 & 1 \end{pmatrix}$.

Transpose and inverse

For invertible A, $(A^{-1})^T = (A^T)^{-1}$.

From $AA^{-1} = I$, transpose both sides: $(A^{-1})^T A^T = I$. Similarly $A^T(A^{-1})^T = I$.

Symmetric matrices

A is symmetric if $A^T = A$. Example: $S = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$.

Its entries satisfy $s_{ij} = s_{ji}$. Products $A^T A$ are always symmetric.

Skew-symmetric matrices

$K^T = -K$. Example $K = \begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}$.

In real skew-symmetric matrices the diagonal entries are zero, since $k_{ii} = -k_{ii}$.

Decompose any square matrix

$$A = \underbrace{\frac{A + A^T}{2}}_{S} + \underbrace{\frac{A - A^T}{2}}_{K}$$

$S^T = S$, $K^T = -K$. Example $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}$: $S = \begin{pmatrix} 1 & 3 \\ 3 & 3 \end{pmatrix}$ and $K = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Why $A^T A$ matters

$(A^T A)^T = A^T A$. For any x, $x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2 \geq 0$.

Example $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ gives $A^T A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$. This anticipates least squares.

Dot product and outer product

For columns $u, v \in \mathbb{R}^n$, $u^T v$ is a scalar dot product, whereas uv^T is an $n \times n$ matrix.

$u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $v = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$: $u^T v = 11$ but $uv^T = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}$.

Permutation matrix

A permutation matrix is obtained by reordering rows of I. Exactly one 1 occurs in each row and column.

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \text{ maps } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \text{ to } \begin{pmatrix} x_2 \\ x_3 \\ x_1 \end{pmatrix}.$$

Left multiplication: rows

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \; P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

$PA = \begin{pmatrix} 3 & 4 \\ 5 & 6 \\ 1 & 2 \end{pmatrix}$. Left multiplication reorders rows.

Right multiplication: columns

$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, $Q = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$.

$AQ = \begin{pmatrix} 2 & 3 & 1 \\ 5 & 6 & 4 \end{pmatrix}$. Right multiplication reorders columns.

Inverse of a permutation

$P^{-1} = P^T$ because $P^T P = I$. The columns are distinct coordinate unit vectors, so their dot products form I.

Consequences: $\|Px\| = \|x\|$, $\det P = \pm 1$.

Determinant sign and parity

One row exchange changes determinant sign. $P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has $\det P = -1$.

Thus $\det(PA) = -\det A$. Two swaps restore the original sign.

Pivoting and $PA = LU$

If a pivot is zero, interchange rows. Partial pivoting chooses the largest absolute candidate in the current column.

For $A = \begin{pmatrix} 10^{-5} & 1 \\ 1 & 1 \end{pmatrix}$, swapping avoids the huge multiplier 10^5.

Record the exchanges in P, giving $PA = LU$.

Matrix columns and combinations

If $A = [a_1 \ \cdots \ a_n]$, then $Ax = x_1 a_1 + \cdots + x_n a_n$.

The set of all such combinations is the column space $\mathrm{Col}(A)$.

What is a vector space?

A vector space over $\mathbb{R}$ has addition and scalar multiplication satisfying the vector-space axioms: zero, additive inverses, associativity, distributivity, and so on.

Examples: $\mathbb{R}^n$, real matrices, polynomials, continuous functions.

The subspace test

$W \subseteq V$ is a subspace if $0 \in W$, $u + v \in W$ for $u, v \in W$, and $cu \in W$ for $u \in W$.

Equivalent: nonempty W is closed under $au + bv$ for all scalars a, b.

Example: line through origin

$W = \{(t, 2t) : t \in \mathbb{R}\} = \text{span}\{(1, 2)\}$.

For $u = (s, 2s)$ and $v = (t, 2t)$, $au + bv = (as + bt, 2(as + bt)) \in W$. Therefore W is a subspace.

Counterexample: shifted line

$S = \{(x, y) : y = 2x + 1\}$. The zero vector does not satisfy the defining equation.

Therefore S is not a subspace. A linear subspace must pass through the origin.

Example: plane through origin

$W = \{(x, y, z) : x + y + z = 0\}$.

Since $z = -x - y$, $(x, y, z) = x(1, 0, -1) + y(0, 1, -1)$.

Thus $W = \text{span}\{(1, 0, -1), (0, 1, -1)\}$.

Span is a subspace

$\text{span}\{v_1, \ldots, v_k\} = \{\sum_i c_i v_i : c_i \in \mathbb{R}\}$.

If $u = \sum_i a_i v_i$ and $v = \sum_i b_i v_i$, then $su + tv = \sum_i (sa_i + tb_i) v_i$ is still in the span.

Membership in a span

Is $(3,5)$ in $\text{span}\{(1,1),(1,-1)\}$? Solve $c_1 + c_2 = 3$, $c_1 - c_2 = 5$.

Adding gives $2c_1 = 8$, so $c_1 = 4$, $c_2 = -1$. Yes.

Column space

$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 0 & 0 \end{pmatrix}$. Then $\mathrm{Col}(A) = \mathrm{span}\{(1,2,0),(2,4,0)\} = \mathrm{span}\{(1,2,0)\}$.

$Ax = b$ is solvable precisely when $b \in \mathrm{Col}(A)$.

Null space

$\text{Null}(A) = \{x : Ax = 0\}$ is a subspace: if $Au = Av = 0$, then $A(su + tv) = 0$.

For $A = \begin{pmatrix} 1 & 2 \end{pmatrix}$, $x_1 = -2x_2$, so $\text{Null}(A) = \text{span}\{(-2, 1)\}$.

Homogeneous versus nonhomogeneous

The solution set of $Ax = 0$ is always a subspace. The solution set of $Ax = b$ with $b \neq 0$ cannot contain 0, hence is not a subspace.

When solvable, $Ax = b$ gives an affine translate $x = x_p + x_n$, $x_n \in \text{Null}(A)$.

Intersection and union

The intersection $U \cap W$ of subspaces is a subspace: it inherits zero and closure from both.

The union need not be: the x-axis and y-axis are subspaces, but $(1, 0) + (0, 1)$ lies in neither axis.

Matrices and polynomials as vectors

Symmetric 2×2 matrices form a subspace of all 2×2 matrices: sums and scalar multiples remain symmetric.

Polynomials $p(t)$ with $p(0) = 0$ form a subspace, but those with $p(0) = 1$ do not.

Geometric summary

In $\mathbb{R}^3$, subspaces are $\{0\}$, lines through the origin, planes through the origin, and all of $\mathbb{R}^3$.

A plane $x + y + z = 1$ is not a subspace; $x + y + z = 0$ is.

Preparation for Week 8 midterm

Be able to: compute A^T and $(AB)^T$; use permutation matrices; state and apply $PA = LU$; test closure under combinations; find spans and column-space membership.

Next: consolidate earlier weeks and practice midterm-style problems.